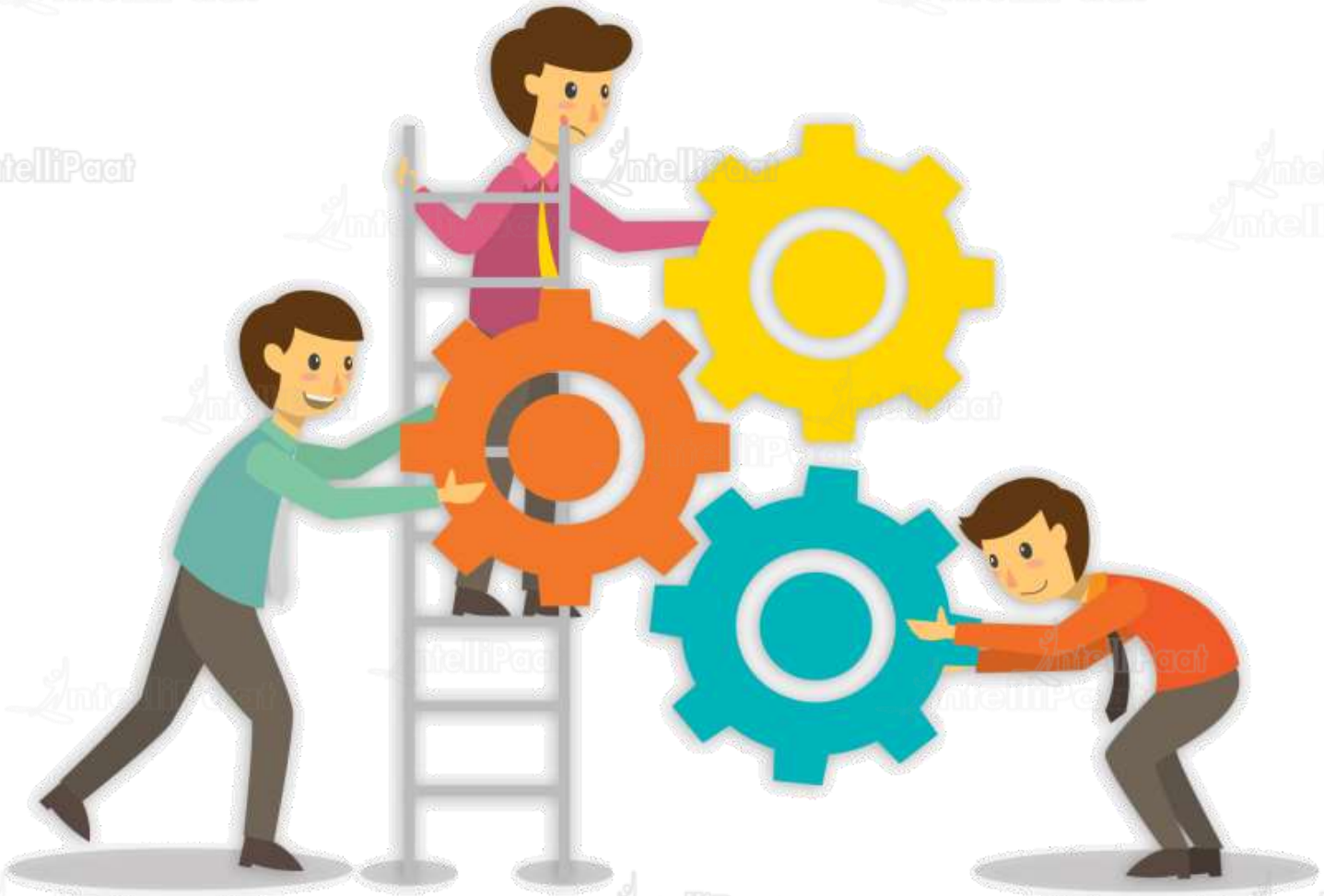
# Python Training

## Python Packages



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# Python Packages

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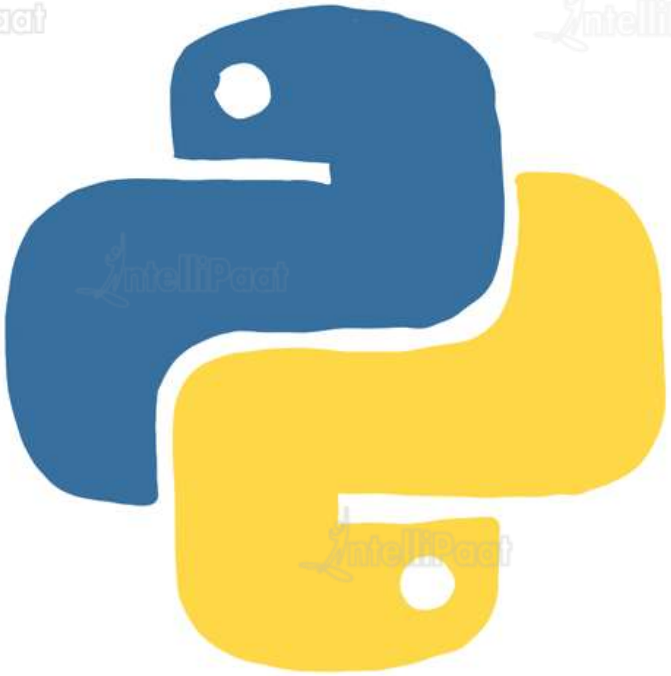
# Python Packages

- Packages are a way of structuring many packages and modules which helps in a well-organized hierarchy of data set, making the directories and modules easy to access.
- Just like there are different drives and folders in an OS to help us store files, similarly packages help us in storing other sub-packages and modules, so that it can be used by the user when necessary.
- To tell Python that a particular directory is a package, we create a file named __init__.py inside it and then it is considered as a package and we may create other modules and sub-packages within it.



Packages

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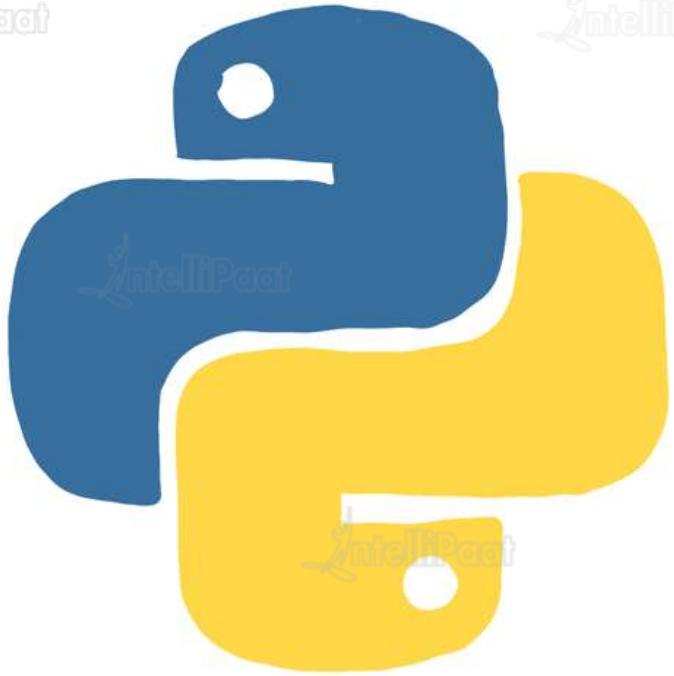
# Python Packages

- This __init__.py file can be left blank or can be coded with the initialization code for the package.
- To create a package in Python, we need to follow these three simple steps:
- First, we create a directory and give it a package name, preferably related to its operation.
- Then we put the classes and the required functions in it.
- Finally we create an __init__.py file inside the directory, to let Python know that the directory is a package.



Packages

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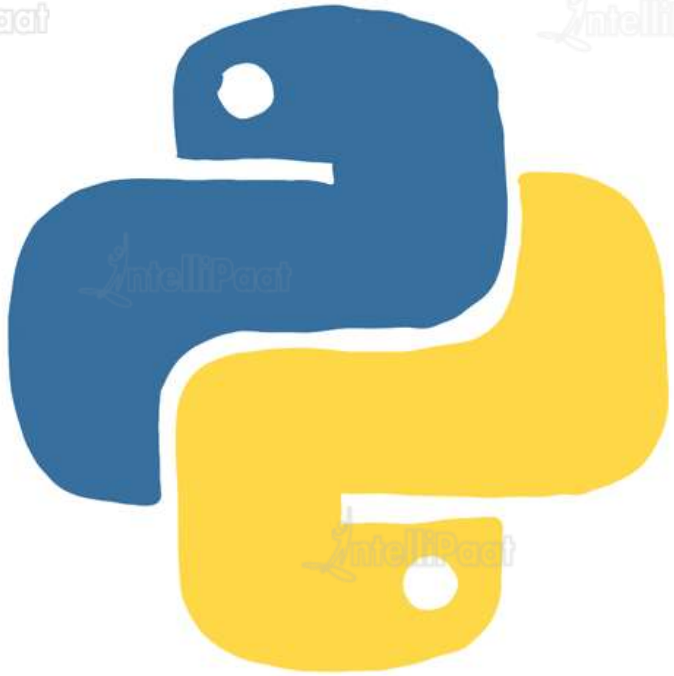
# Python Packages

- Suppose you have developed a very large application that includes many modules.
- As the number of modules grows, it becomes difficult to keep track of them all if they are dumped into one location.
- This is particularly so if they have similar names or functionality.
- You might wish for a means of grouping and organizing them.
- Packages allow for a hierarchical structuring of the module namespace using dot notation.
- In the same way that modules help avoid collisions between global variable names, packages help avoid collisions between module names.



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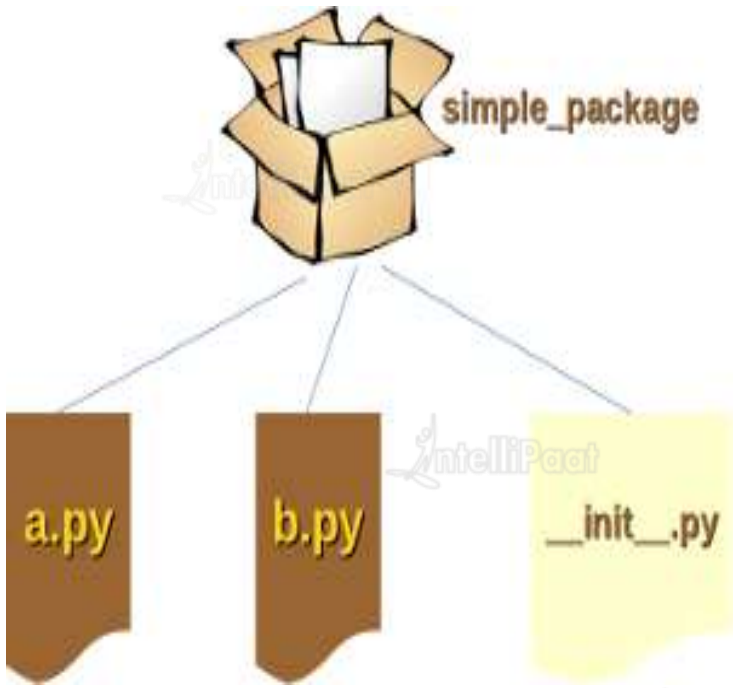
# Python Packages

- We will demonstrate with a very simple example how to create a package with some Python modules.
- First of all, we need a directory. The name of this directory will be the name of the package, which we want to create.
- We will call our package "simple_package". This directory needs to contain a file with the name "__init__.py".
- This file can be empty, or it can contain valid Python code.
- This code will be executed when a package will be imported, so it can be used to initialize a package, e.g. to make sure that some other modules are imported or some values set.

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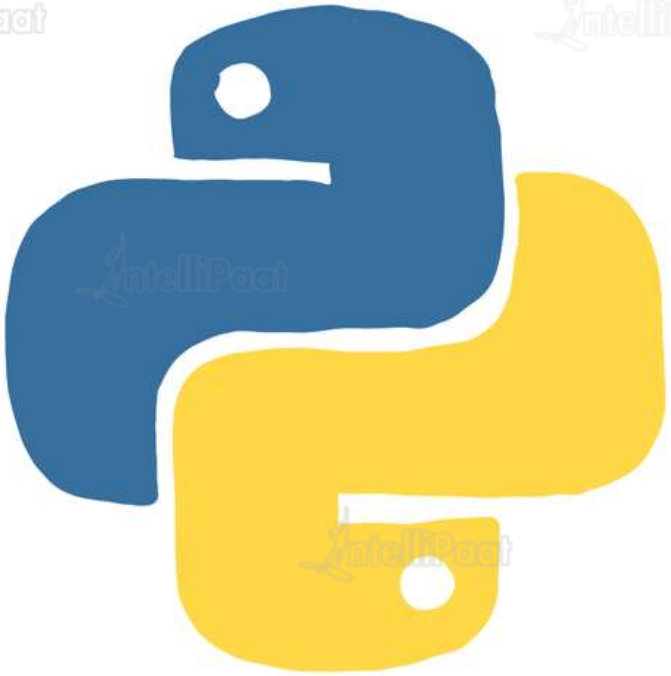
# Various ways of accessing the Packages

- Suppose the cars and the brand directories are packages.
- For them to be a package they all must contain __init__.py file in them, either blank or with some initialization code.
- Let’s assume that all the models of the cars to be modules.
- Use of packages helps importing any modules, individually or whole.
- Suppose we want to get Bmw i8. The syntax for that would be:
- 'import' Cars.Bmw.x5
- While importing a package or sub packages or modules, Python searches the whole tree of directories. looking for the particular package and proceeds systematically as programmed by the dot operator.



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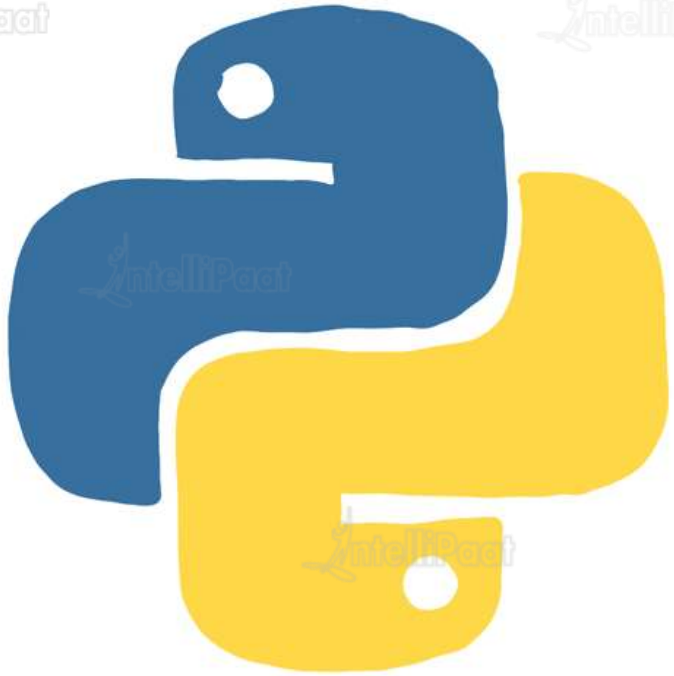
# ‘From…Import’ in Packages

- Now, whenever we require using such function we would need to write the whole long line after importing the parent package.

- To get through this in a simpler way we use ‘from’ keyword.

- For this we first need to bring in the module using ‘from’ and ‘import’:
  from Cars.Audi import a8
- Now we can call the function anywhere using
  a8.get_buy(1)
- There’s also another way which is less lengthy. We can directly import the function and use it wherever necessary. First import it using:
  from Cars.Audi.a8 import get_buy
- Now call the function from anywhere:
  get_buy(1)



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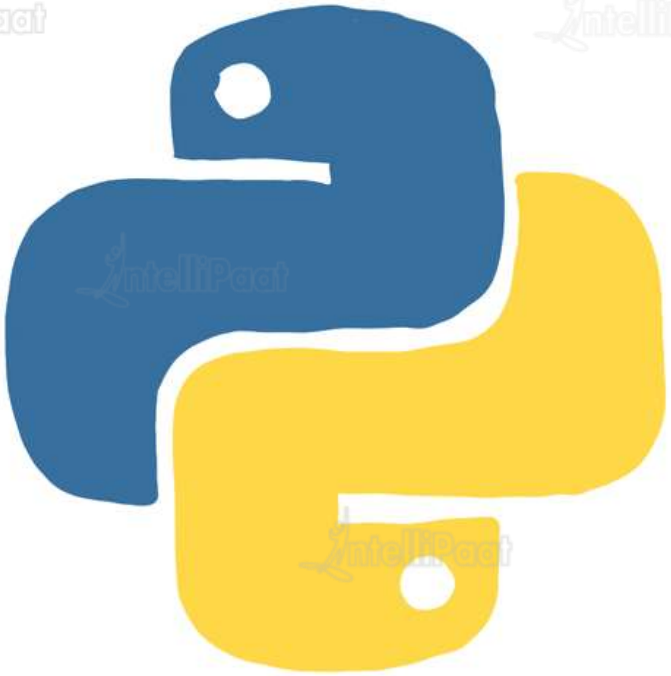
# ‘From…import*’ in Packages

- While using the from…import syntax, we can import anything from submodules to class or function or variable, defined in the same module.
- If the mentioned attribute in the import part is not defined in the package then the compiler throws an ImportError exception.
- Importing sub-modules might cause unwanted side-effects that happens while importing submodules explicitly. Thus we can import various modules at a single time using * syntax.
- The syntax is:
  Impofrom Cars.Chevrolet import *
- This will import everything i.e., modules, sub-modules, function, classes, from the sub-package.
- While using just the import syntax, one must keep in mind that the last attribute must be a subpackage or a module, it should not be any function or class name.



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